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BSc Cyber Security

Task 1: Sorting Algorithms

Input Array: The input array based on the student ID (ALI22593481) is:

[2, 2, 5, 9, 3, 4, 8, 1]

(a) Merge Sort

Steps for merge sort:

- Split Phase: Recursively divide the array into halves until single-element arrays are achieved.

[2, 2, 5, 9, 3, 4, 8, 1]

[2, 2, 5, 9] [3, 4, 8, 1]

[2, 2] [5, 9] [3, 4] [8, 1]

[2] [2] [5] [9] [3] [4] [8] [1]

2. Merge Phase: Combines arrays in sorted order.

[2] + [2] = [2, 2]

[5] + [9] = [5, 9]

[3] + [4] = [3, 4]

[8] + [1] = [1, 8]

[2, 2] + [5, 9] = [2, 2, 5, 9]

[3, 4] + [1, 8] = [1, 3, 4, 8]

[2, 2, 5, 9] + [1, 3, 4, 8] = [1, 2, 2, 3, 4, 5, 8, 9]

Sorted Output: [1, 2, 2, 3, 4, 5, 8, 9]

(b) Quick Sort

Steps for quick sort:

- Use the last element as the pivot: Partition the array such that the elements less than the pivot are on the left and greater on the right.

Initial array: [2, 2, 4, 9, 3, 4, 8, 1]

Pivot = 1: [1] [2, 2, 5, 9, 3, 4, 8]

Pivot = 8: [1, 2, 2, 5, 3, 4] [8, 9]

Pivot = 4: [1, 2, 2, 3] [4, 5] [8, 9]

Combine: [1, 2, 2, 3, 4, 5, 8, 9]

Sorted Output: [1, 2, 2, 3, 4, 5, 8, 9]

(c) Insertion Sort

Iteratively build a sorted array by picking elements and placing them in the correct position.

1. Start with [2].
2. Insert 2: [2, 2]
3. Insert 5: [2, 2, 5]
4. Insert 9: [2, 2, 5, 9]
5. Insert 3: [2, 2, 3, 5, 9]
6. Insert 4: [2, 2, 3, 4, 5, 9]
7. Insert 8: [2, 2, 3, 4, 5, 8, 9]
8. Insert 1: [1, 2, 2, 3, 4, 5, 8, 9]

Sorted Output: [1, 2, 2, 3, 4, 5, 8, 9]

Task 2: Dijkstra's Algorithm

According to the instructions, following are the weight of the edges (D1.....D9) according to my student ID (ALI22593481). As there are 8 digits in the student ID so the ninth digit is 1.

D1-----2

D2-----2

D3-----5

D4-----9

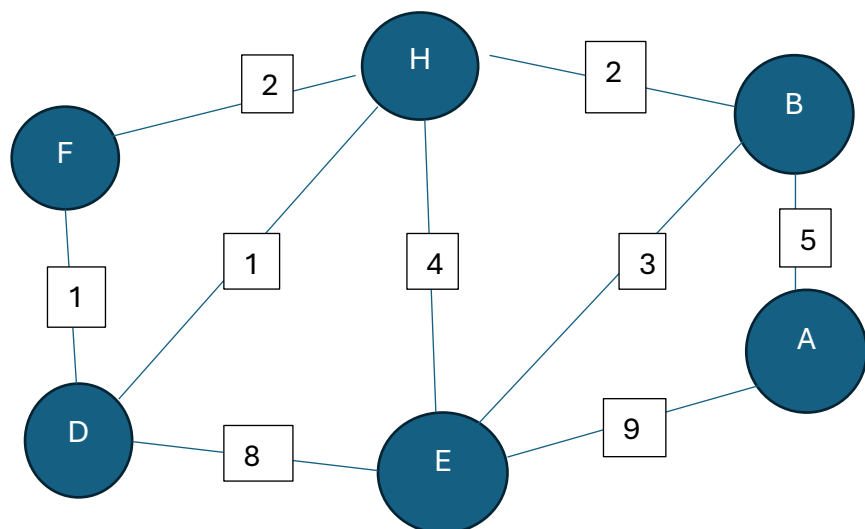
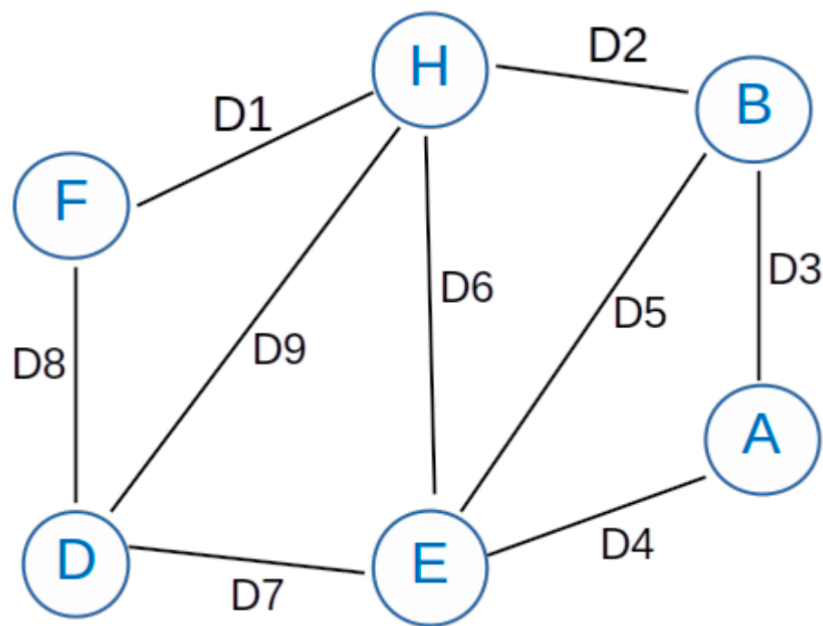
D5-----3

D6-----4

D7-----8

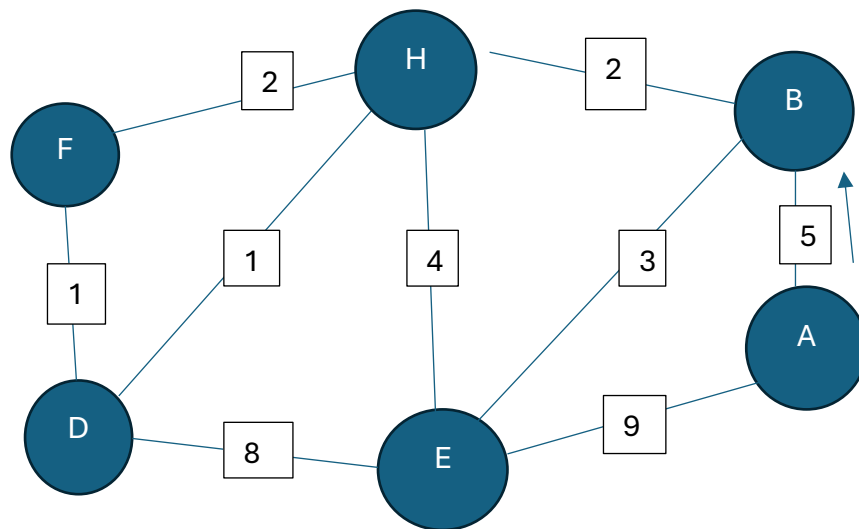
D8-----9

D8-----1

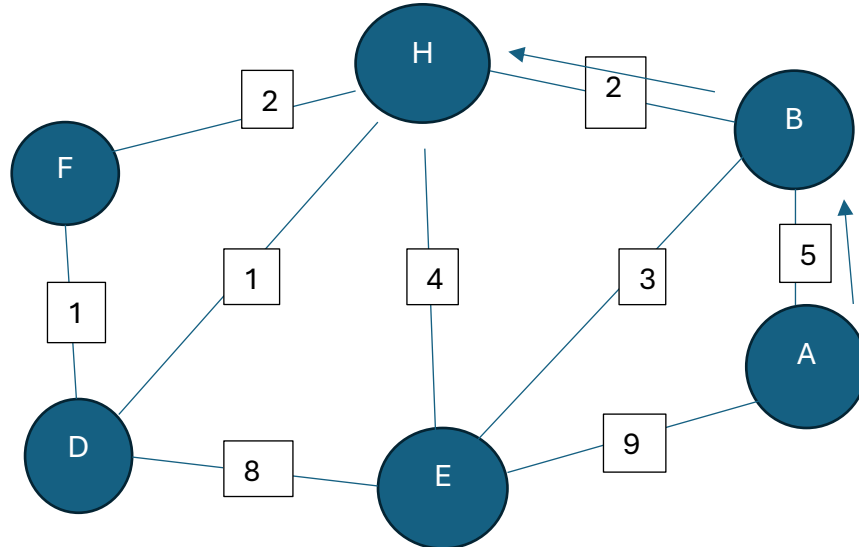


Shortest paths from Node A:

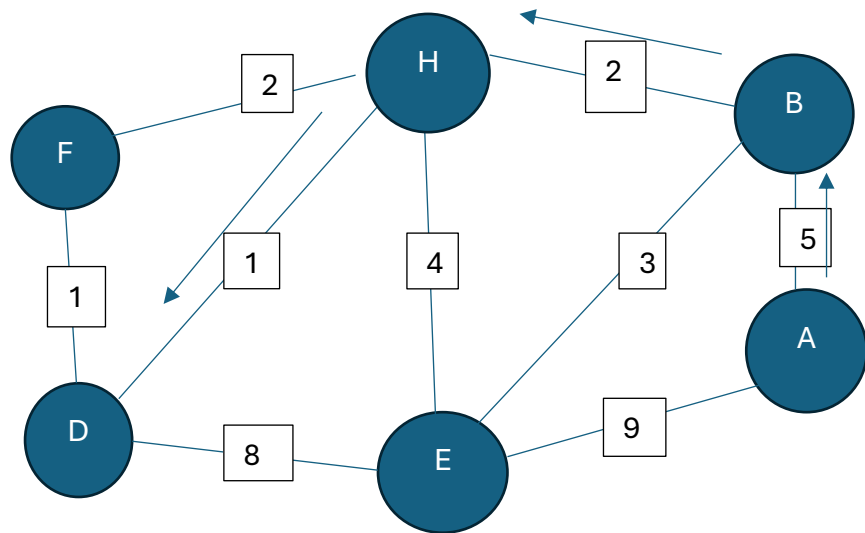
- $A \rightarrow B$: Distance = 5



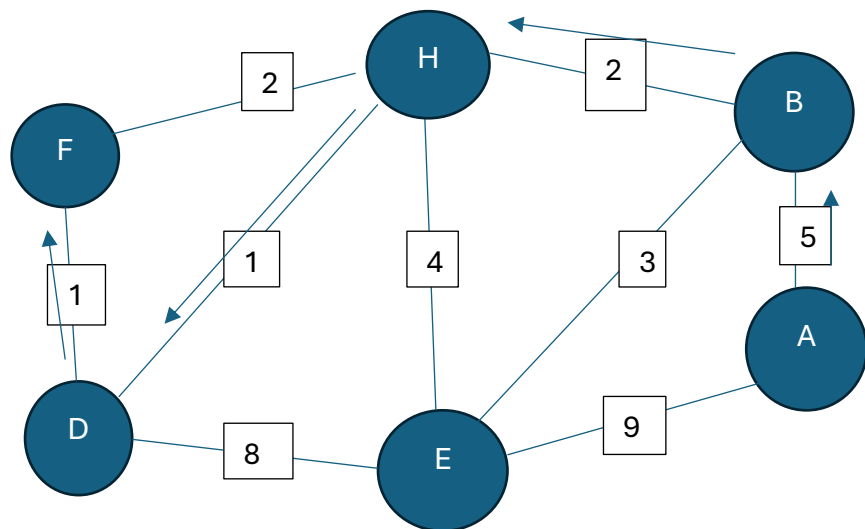
- $A \rightarrow H$: Distance = 7(via B)
Update H as 7 as it went via B which is 5.



- $A \rightarrow D$: Distance = 8 (via B and H)
Distance from A to D is 8 as we went to shortest way from H (7) which is H to D.

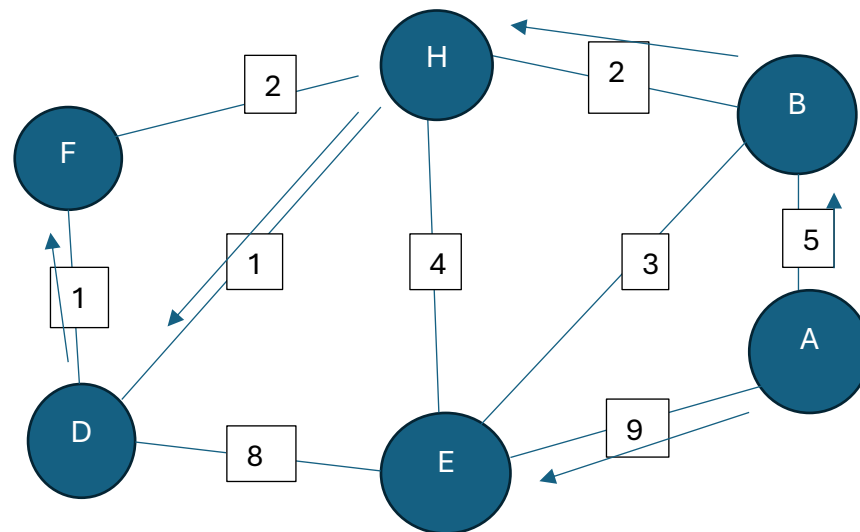


- $A \rightarrow F$: Distance : 9(via B H and D)
As we were on node D (8) and there is the shortest distance 1 from D to F



- $A \rightarrow E$: Distance : 9

As we were on node F and there is no other way to go on E so we came back to node D but still the distance from node d is E is going to be the large. As per the rule we are following the shortest distance so the shortest one is straight from A to E.



Task 3: Hash Table and Binary Search Tree

Input name : Muhammad Awais Ali

(a) Hash Table:

- Each letter is hashed to its index in the English Alphabet.
- Table size = 26 (letters a to z). Hashing steps with Chaining:

1	a	→ a, a, a, a
2		
3		
4	d	
5		

6		
7		
8	h	
9	i	→ i
10		
11		
12	l	
13	m	→ m,m
14		
15		
16		
17		
18		
19	s	
20		
21	u	
22		
23	w	
24		
25		
26		

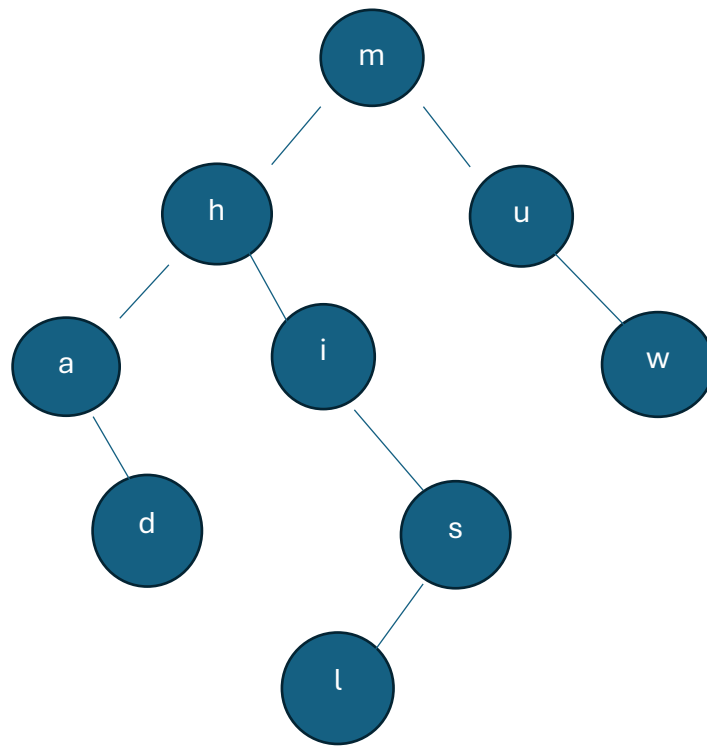
Final Hash Table:

Index	Values
1	a, a, a, a, a
4	d
8	h
9	l, i
12	l
13	m, m, m
19	s
21	u
23	w

(b) Binary Search Tree:

- Insert the characters of the name “Muhammad Awais Ali” into a BST based on their ASCII values.

- Insertion order : m, u, h, a, m, m, a, d, a, w, a, i, s, a, l, i



Explanation:

- The root node is m and has left child h and right child u.
- The h node has left child a and right child i.
- The a node has only right child d, i node has 1 right child s, and u node is also having 1 right child w.
- On node s there is only one left child l and that covers all the characters of name.

Thanks!